

[Collaborative Docs](#) >

Discussion Agenda 03/25/2014

Below is a discussion agenda / live progress report in an approximate order of importance for the 5th Web Conference of the AGORA Project (Mar. 25, 2014, 10 am PDT/6 pm CET). Members of the AGORA Steering Committee (Abel, Gnedin, Madau, Mayer, Primack, Teyssier, Wadsley) + Hahn, Turk, Kim discussed some of these items on Feb. 6, 2014. **Speakers' names in the conference, and important updates are in red color.**

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[1. Progress Report \(as of Mar. 2014\)](#)

We are actively making a lot of progress, chiefly, in three different fronts below (numbered tentatively). If you wish to contribute to any of these efforts, please talk to the respective leaders.

	PAPER "2"	PAPER "3"	PAPER "4"																																																
GOALS	• Run Proof-of-concept-DM (~1e11 Ms halo)	• Implement and test common astrophysics package • Run Proof-of-concept-NOSUBGRID (~1e11 Ms halo)	• Calibrate SF/Fbck recipes using common disk ICs																																																
LEADERS	• Oliver Hahn (ETH)	• Nick Gnedin (Fermilab) • Piero Madau (UCSC) • Britton Smith (Edinburgh)	• Oscar Agertz (Surrey) • Romain Teyssier (Zurich) • James Wadsley (McMaster)																																																
PARTICIPANTS	• Working Groups engaged: – III (common cosmo ICs) – IV (common analysis) – VI (dwarf galaxies) • People in charge:	• Working Groups engaged: – I (common physics) – III (common cosmo ICs) • People in charge:	• Working Groups engaged: – II (common disk ICs) – IV (common analysis) – V (subgrid physics) • People in charge:																																																
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DELIVERABLES IN 2014	• Push out PAPER "2"	• Push out PAPER "3"	• Push out PAPER "4"																																																
	• Thanks to the great effort	• Thanks to the great effort by	• Core participating codes have																																																

PROGRESS MADE OR BEING MADE	by Oliver Hahn, a new parameter set of the target halo IC is ready. Please see Paper 2's Workspace and the MUSIC v1.5 release note for more information. You can find the IC used for Paper 2 there as well. • Gadget (Hahn), Pkdgrav2 (Guedes) and ART-II (Leitner) groups are performing initial tests of this new IC. For example, high-res runs using the new IC are already completed by Gadget (Hahn) and Pkdgrav2 (Guedes) groups. No serious contamination is found inside the target halo at $z=0$. • The Enzo code is being tested for its new capabilities (thanks to Simpson) to deal with the new ellipsoidal ICs. • An email will be sent out to initiate low-res runs comparison. We will then set up a telecon to discuss the strategy to perform and analyze high-res runs.	Britton Smith, a fully tabulated equilibrium mode of the GRACKLE cooling library is now ready. Please see his announcements (1, 2) for more information. • Gadget (Choi, Nagamine, Rocha, Onorbe, et al.), ART-II (Leitner et al.), and Enzo (Smith, Turk, Butler, et al.) groups have implemented and started testing one or both modes of the GRACKLE cooling. Efforts by ART-I (Ceverino et al.), Gasoline (Shen, Wadsley et al.), Ramses (Teyssier et al.) groups are underway. • A discussion on the API change is underway. Additionally, the Gadget-UNLV group reports a small difference in cooling rates between GRACKLE and the traditional routine. • These issues plus how to move Paper "3" forward will be discussed in the upcoming telecon among Paper "3" participants.	been identified (3 AMRs + 4 SPHs). Thanks to the great effort by Oscar Agertz, Romain Teyssier and James Wadsley, "actual" comparison has been initiated first with cooling+star formation but with no feedback. Please see Paper 4's Workspace for more information. • Consensus to be made for the GRACKLE cooling and feedback recipes.
TIMELINE IN 2014	• (to be filled)	• (to be filled)	• (to be filled)
SUBSCRIBE TO PROGRESS	• Please see the progress report in Paper 2's Workspace and/or in WG III's Workspace. • MUSIC Google group	• Please see the progress report in Paper 3's Workspace. • GRACKLE Google group • GRACKLE download & documentation	• Please see the progress report in Paper 4's Workspace and/or in WG V's Workspace. • yt-AGORA Google group

[2. yt-AGORA mini-workshop @UCSC \(Mar. 23-25, 2014\)](#)

(1) Workshop Overview:

Thanks to the great effort by Matthew Turk et al., a 3-day yt-AGORA mini-workshop was held at UC Santa Cruz from Mar. 23-25, 2014

([announcement](#), [logistics](#)). Organized by Matthew Turk, Miguel Rocha, and Alex Bogert. 19 participants on-site, 3 participants online. Right after the UC-HiPACC conference [Computational Astrophysics: Toward Exascale](#) at LBNL. A [summary email](#) was sent out to the yt-AGORA mailing list.

(2) Participants have focused on:

- Better Rockstar integration, potentially with SAMs / Halo analysis redesign
 - S. Skillman, B. Smith, M. Turk, M. Rocha, C. Hummels, D. Silvia, H. Egan, B. Crosby et al.
- SPH smoothing
 - M. Turk, N. Goldbaum, B. Keller, G. Altay et al.
 - B. Keller (McMaster; Topsy) and G. Altay (Georgia Tech; Gadget-OWLS) have made important progress.
 - The improvement on the yt-SPH (as of Feb. 2014) is demonstrated in Gadget3-UNLV [images](#) for Paper 4.
- GPU volume rendering / Interactive visualization via yt+IPython notebook (a.k.a. PyRGBA, Python Rendering and GPU Bitmap API)
 - A. Bogert, M. Turk, S. Skillman, J. Holdener, N. Smith et al.
 - A [live demo](#) by Ji-hoon Kim, Matthew Turk and Alex Bogert via SeeVogh's screenshare.
 - See [this page](#) for more information.
- Scalability of octree datasets / Frontend maintenance
 - M. Turk, K. Arraki, A. Elvin et al.
- Many more topics on [Trello](#), Some pictures taken at the workshop (Mar. 23): [1](#), [2](#), [3](#)

3. NERSC Allocations and Strategy for Data Storage

(1) NERSC Allocations:

- [Joel Primack](#) and Piero Madau are pleased to announce that the AGORA Collaboration Database has received an Allocation Year 2014 Data Pilot award from National Energy Research Scientific Computing (NERSC). The allocation includes 5M cpu-hours (Cray XT4-equivalent MPP hours for data analysis, mainly with yt), and 0.6M storage resource units (SRUs) enough to transfer ~100 TB into and out of their HPSS, starting Jan. 14, 2014. This will become the mass storage for AGORA, and the NERSC computer time will allow us to run yt and Sunrise on the simulation outputs. You will be able to acquire user accounts on these resources very soon.
- Additional storage and post-processing compute time may become available at the Hyades/Huawei system at UCSC, the San Diego Supercomputer Center at UCSD, or the Data-Scope system at John Hopkins.

(2) Strategy to Manage and Use NERSC Resources:

- How to organize the resources and their access? Below is an initial proposal by the Steering Committee, as discussed in Feb. 6, 2014, and confirmed in the Web Conference.
 - As many users as needed will become available to the AGORA Collaboration (per conversation between Joel Primack and Richard Gerber/Peter Nugent at NERSC).
 - Organize the storage by code groups ("workspace"). Different versions of the same code will be considered as different code groups (e.g. Gadget-3-UNLV, Gadget-3-SPHS, etc). Each "workspace" is to be managed by two code Czars (a lead Czar + a deputy Czar).
 - [Members of each code group have immediate access to their own "workspace" \(up to 10 TB\), and post and verify the simulation data until they are satisfied. Once the code group is fully satisfied with the simulation outputs, then it opens up the "workspace" by changing the permission and invite others from the science Working Groups.](#)
 - We will soon build a new page here on the Google Project Workspace on how to use the NERSC resources. We will also update the [simulation tracker page](#) to index which simulation data is located where ([example](#)).

- Interactive visualization via yt+IPython notebook is being developed by Alex Bogert, Matthew Turk et al., and will become available for all AGORA participants on NERSC resources. Sunrise will also be installed on the NERSC system by Miguel Rocha et al., yt being the front-end for Sunrise. **All these will provide a complete data storage+analysis pipeline for AGORA collaboration members, although to be restricted to the AGORA-related usage only.**

- Please see [this page](#) to learn how the key technology is being developed. The software (a.k.a. PyRGBA) will be released publicly.
- For this purpose, NERSC has installed one Kepler K20 and one Kepler K40 card in the system. It is considering to purchase more of the latest Nvidia graphics cards with the Kepler architecture, potentially with matching funds from UC–HiPACC (probably very soon; as of Mar. 2014).

(3) Strategy to Store Simulation Outputs:

- As we start to produce simulation data, and NERSC resources become available for the Project, we revisit the time-stepping strategy for data storage. Tentative policies as of Mar. 26, 2014 based in part on the proposal by Peter Behroozi and Joel Primack.
- Each of the participating codes will generate large quantities of unprocessed, intermediate data, in the form of “checkpoints” describing the state of the simulation at a given time. **We recommend total 201 timesteps: the first 21 timesteps equally spaced in redshift, $\Delta z = 0.5$, from $z=20$ to $z=10$, and the remaining 180 timesteps equally spaced in $\log(\text{expansion parameter})$, $\Delta \log(a) = |\log(1/11)/180|$, from $z=10$ to $z=0$. Plus redshift snapshots at $z = 6.0, 3.0, 2.0, 1.0, 0.5, 0.2, 0.0$ at the very least.**
- Each group is advised to make sure that they can restart the simulation from a point of your interest by storing the “checkpoints”, and also that their outputs are “analyzable.” (Because “checkpoints” might not necessarily be “analyzable” in some codes.) This way, one may add more timesteps in the time period of your interest.
- Each group may be advised to store more timesteps depending on the science questions being pursued (e.g. ~ 1000 timesteps to study the orbital motions of satellite galaxies). Each group may also want to store additional outputs at slightly earlier redshift, shifted by $\Delta z \sim 0.05$. This extra information will help us investigate whether an inter-platform offset in time-stepping causes timing differences for halo mergers, for example.

4. Other News

(1) Flagship Paper (Paper "1"):

- The Flagship paper has now been accepted by the ApJS and **published** ([local](#); in the Jan. 2014 issue). The published version of the manuscript is also available on the [arXiv](#).

(2) Mini-workshop Opportunities for Individual Working Groups:

- UC–HiPACC@UCSC and KIPAC@Stanford are willing to provide funding opportunities for working group leaders to organize mini-workshops (~ 10 people) at either place as their science results come in. Feel free to discuss this with the [project coordinator](#) if you wish to organize one.

(3) Workshop in 2014 (preliminary):

- Annual Santa Cruz Galaxy Workshop (Aug. 11–15, 2014) + 3rd AGORA Workshop at UCSC (Aug. 15–17, 2014)

(4) Useful Links:

- Collection of softwares relevant to AGORA effort: [here](#)
- Quick links to the important pages in AGORA Workspace: [here](#)
- Internal policy regarding the disclosure of AGORA simulation results: [here](#)

